



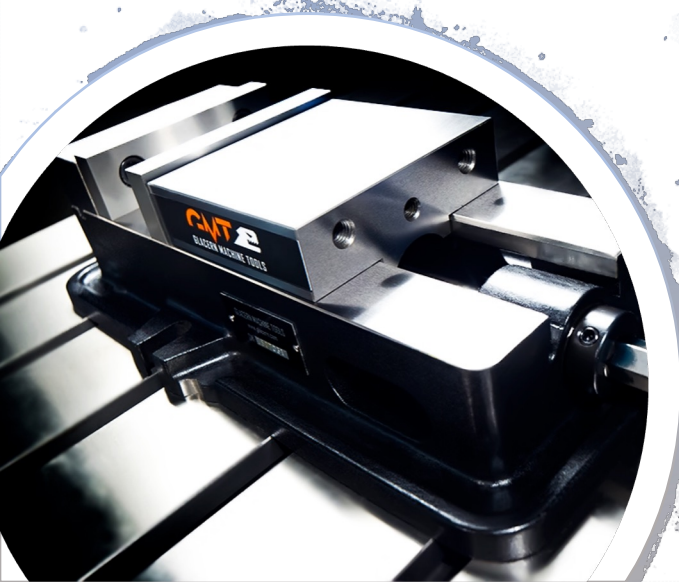
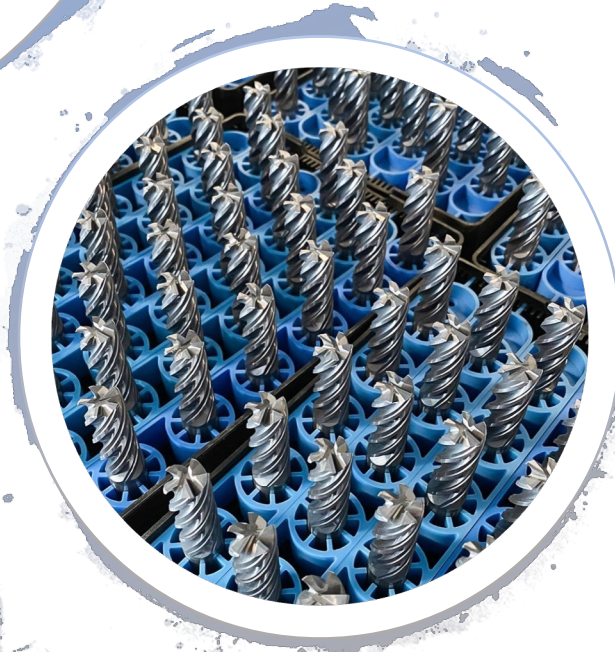
BRUIN 
RACING

Design for
Manufacturability



What Goes into CNC Machining

- Programming
- Tooling
- Workholding



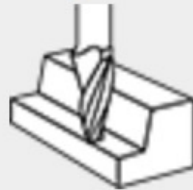
Basics of Tooling



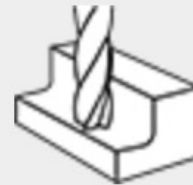
Square



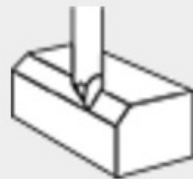
Ball



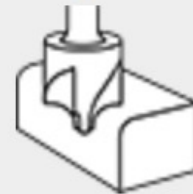
Tapered



Rounded-Edge
Square



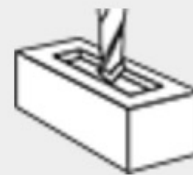
Chamfer



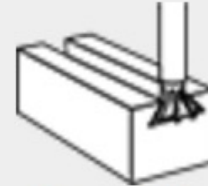
Corner
Rounding



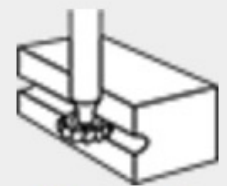
Threading



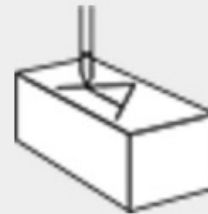
Drill/Mill



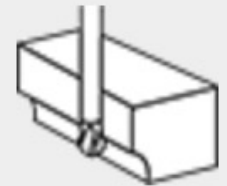
Dovetail



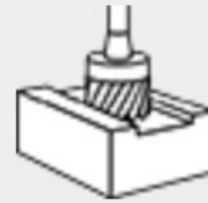
Keyseat



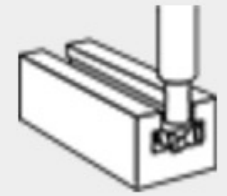
Engraving



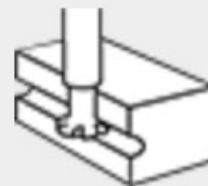
Undercutting



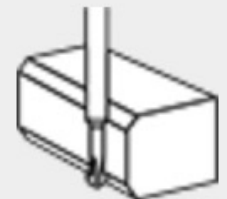
Face Milling



T-Slot

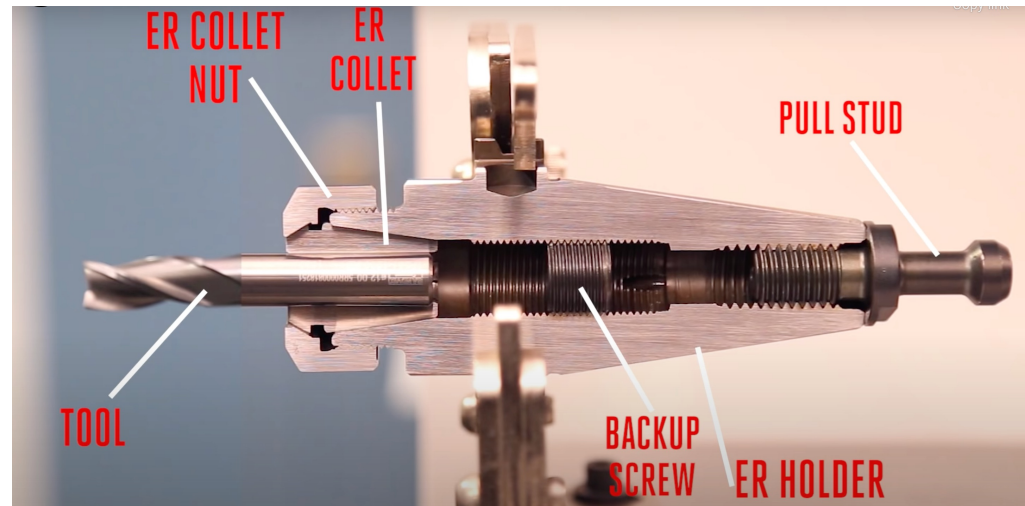


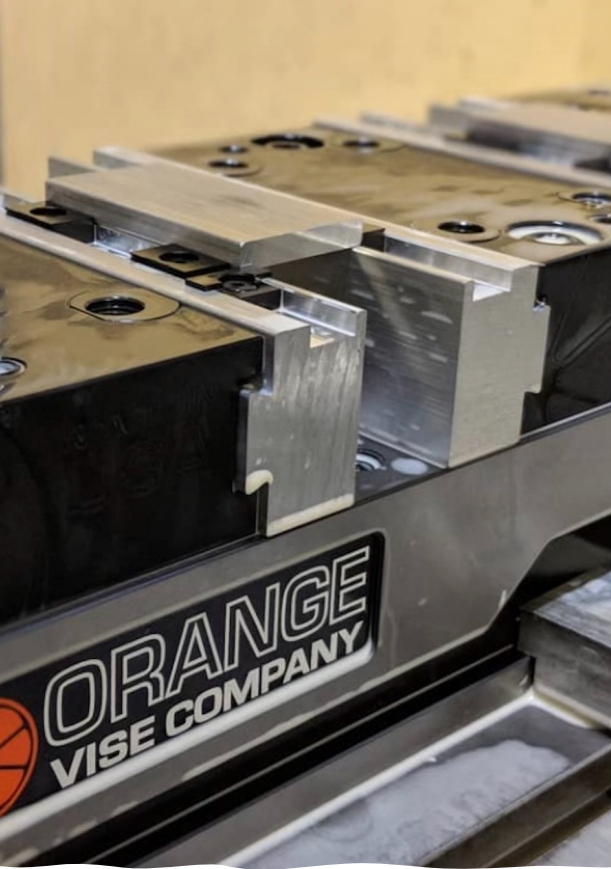
Convex Radius



Double
Chamfer

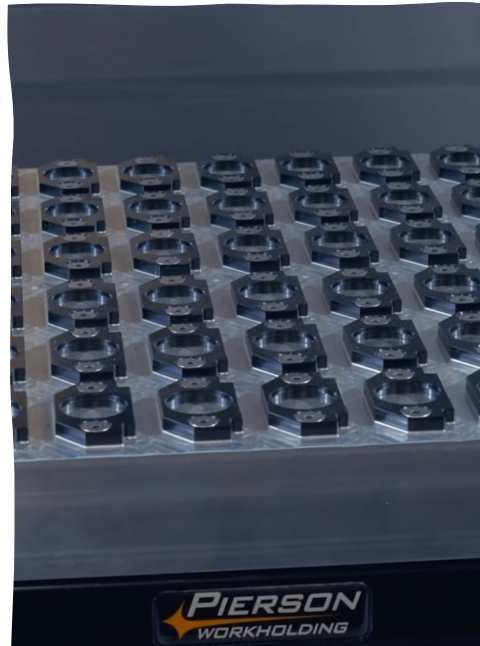
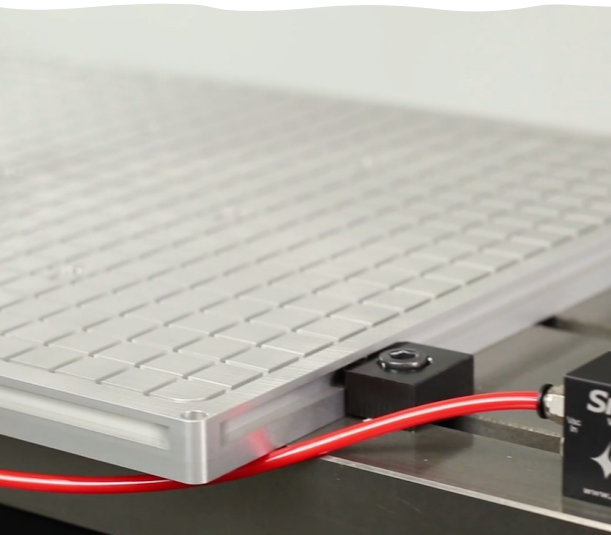
Basics of Tool Holding

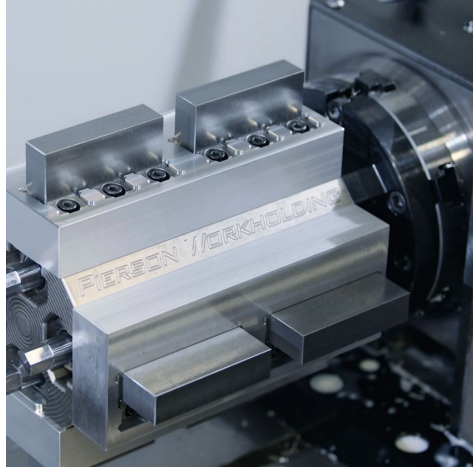




Basics of Workholding

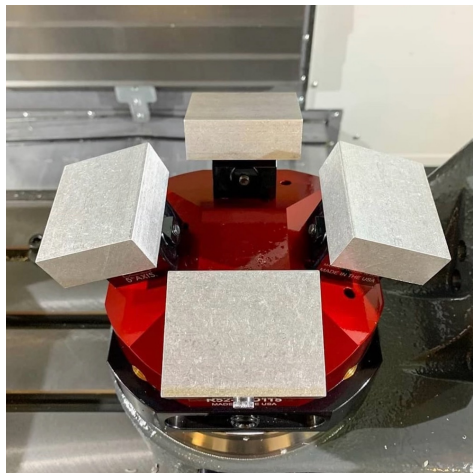
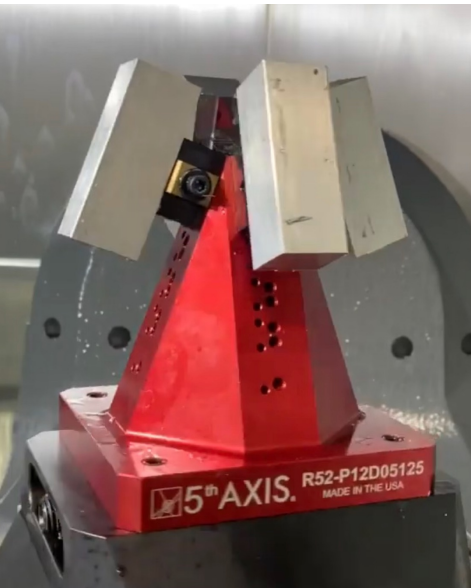
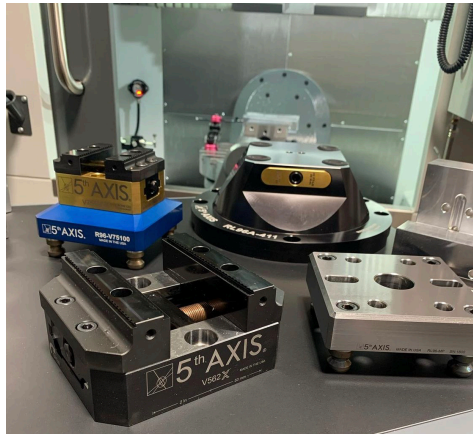
- Low Volume / Prototyping
- High Density / Production





Basics of Workholding

- Complex Geometry (4th / 5th Axis)



Design for Manufacturability

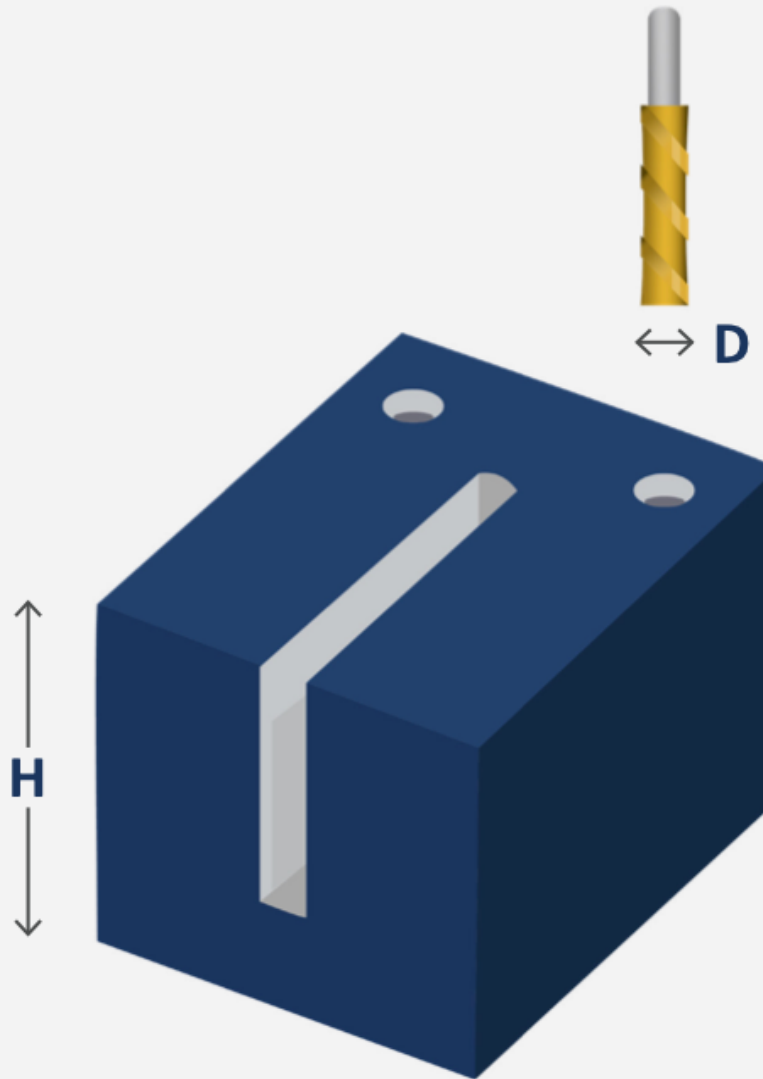
- Part Size
- Material
- Tolerance
- Complexity
- Surface Finish

Basic Cost Formula:

Material + Machining Time* + Profit Margin

* Machining time is a function of material removal rate (Tool diameter, Material Hardness, Machine Speed,)

Deep Narrow Slots



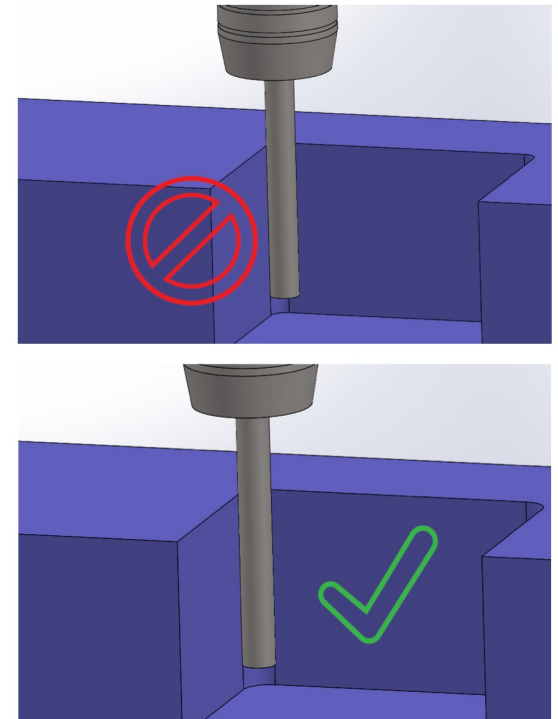
$H < 3D$

D = Diameter of smallest tool needed for final feature

- Try to avoid pockets and slots that are narrow and deep. Longer tools are more prone to breakage and chip removal becomes difficult, especially when the pockets and slots are blind.

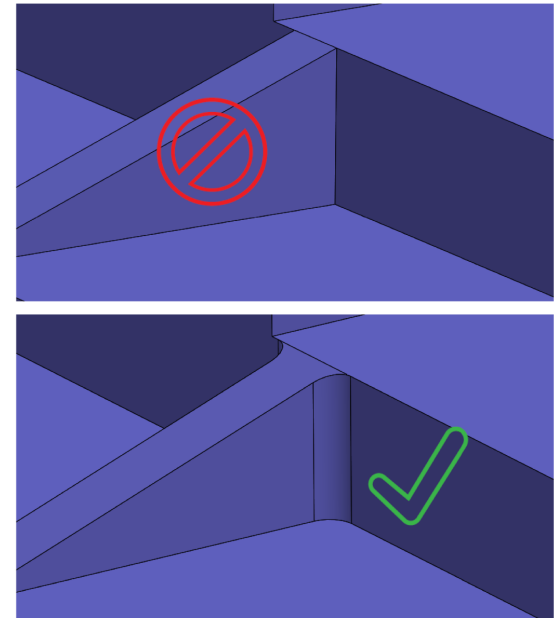
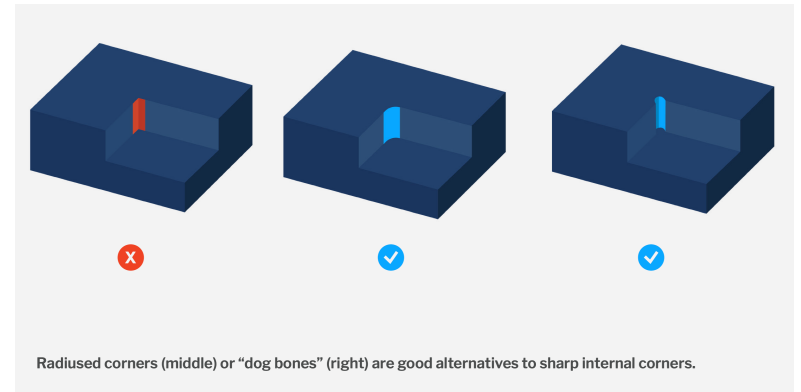
Interior Fillet

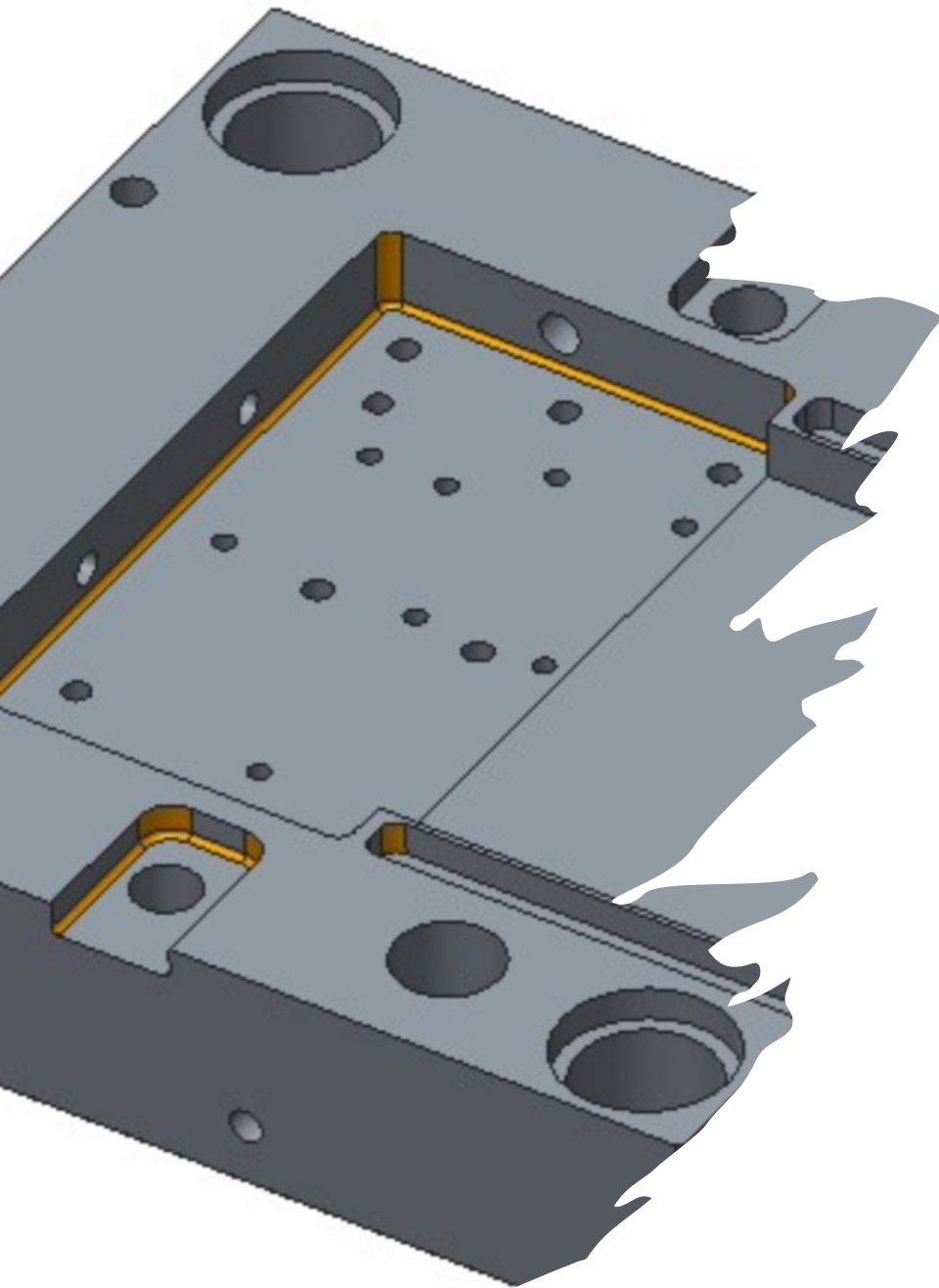
- Interior fillet size less than 3:1 length to diameter ratio requires a small tool to cut. A larger radii uses a larger tool which may result in a better surface finish and tighter tolerances.



Sharp Inside Corner

- Only inside radii or inside corners from a perpendicular direction of the round tool need to have fillets. Avoid sharp square inside corners.



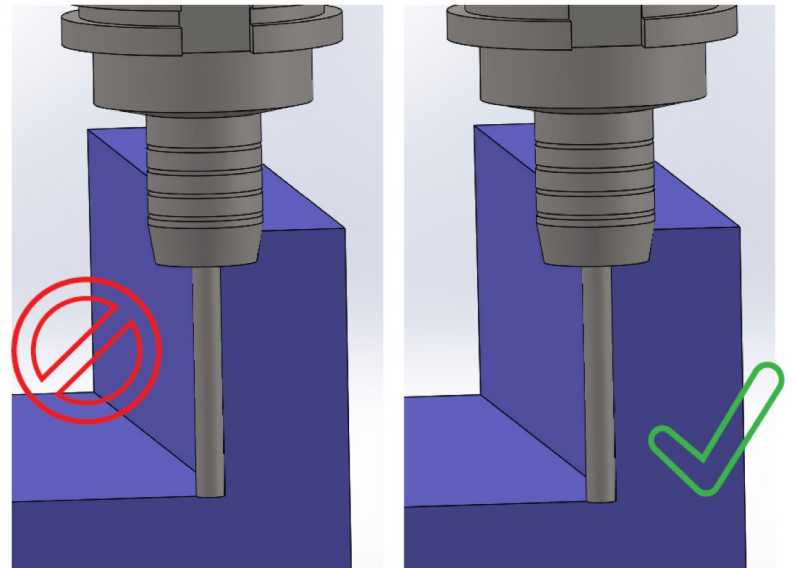


Side Radius and Bottom Radius

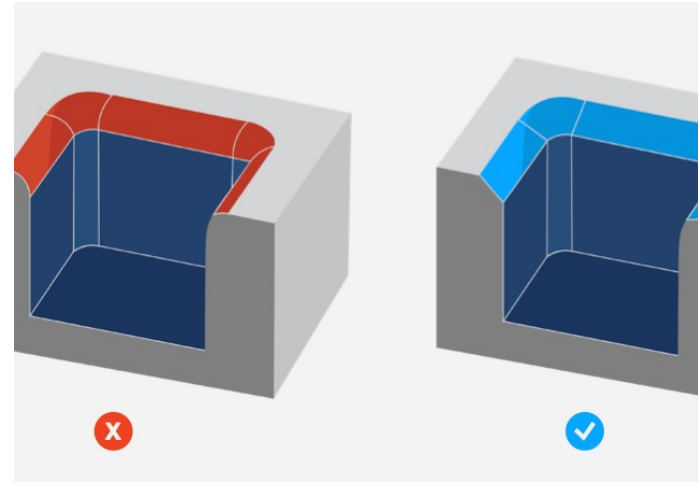
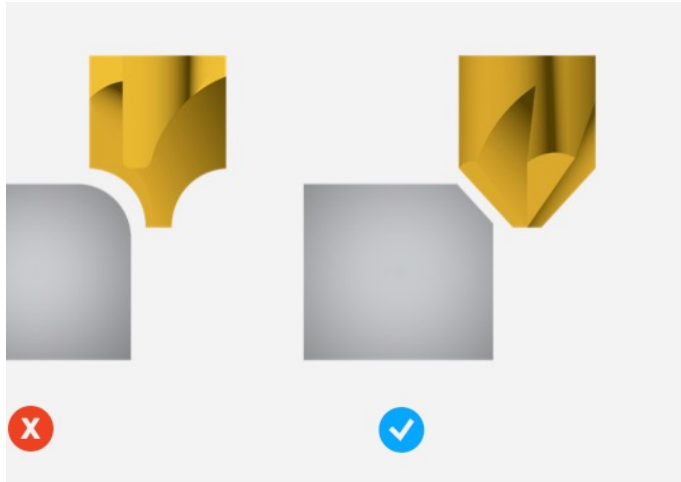
- Use of standard side radius and bottom radius for milling features will ease manufacturing of milling features with standard available milling tools. For reducing, machining cycle time and tool setup cost, it is recommended to avoid non-standard side radius and bottom radius. It is recommended to use single standard side radius and single standard bottom radius.

DEEP POCKETS

- Pockets deeper than 2.00" may require longer tooling which creates chatter (tool vibration) resulting in potential surface finish and tolerance issues. If deep pockets are required and you have clearance, creating a step down will allow the manufacturer to use shorter tooling which will reduce cost.



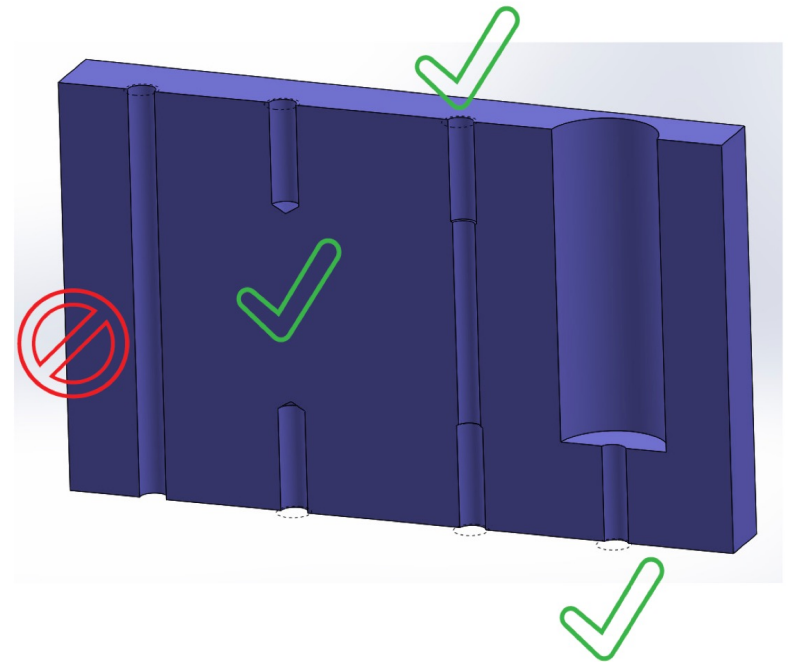
OUTSIDE CORNERS



- For outside corners, chamfers are preferable over radii, as shown below. An outside radius requires a form-relieved cutter and a precise setup, or surface machining of the radius. Both are expensive.

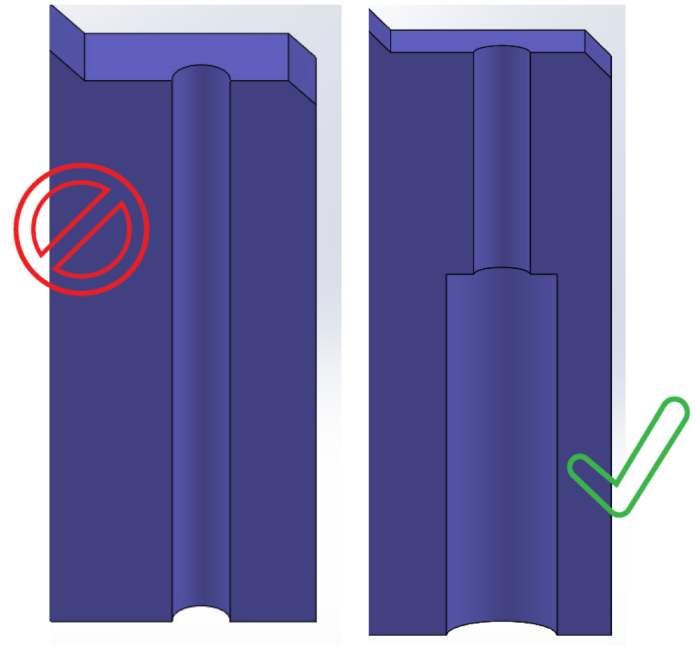
THREADED HOLES

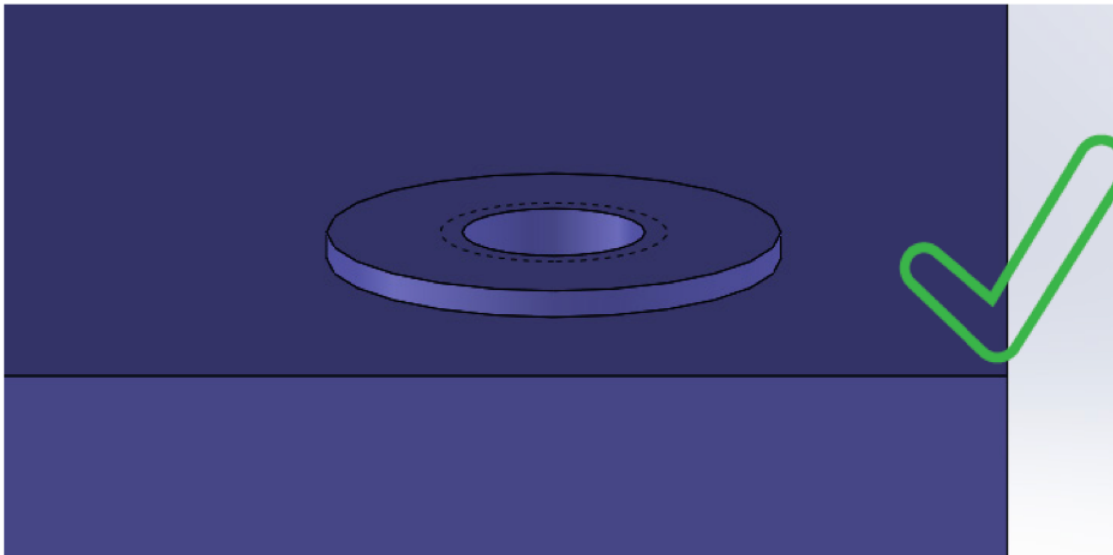
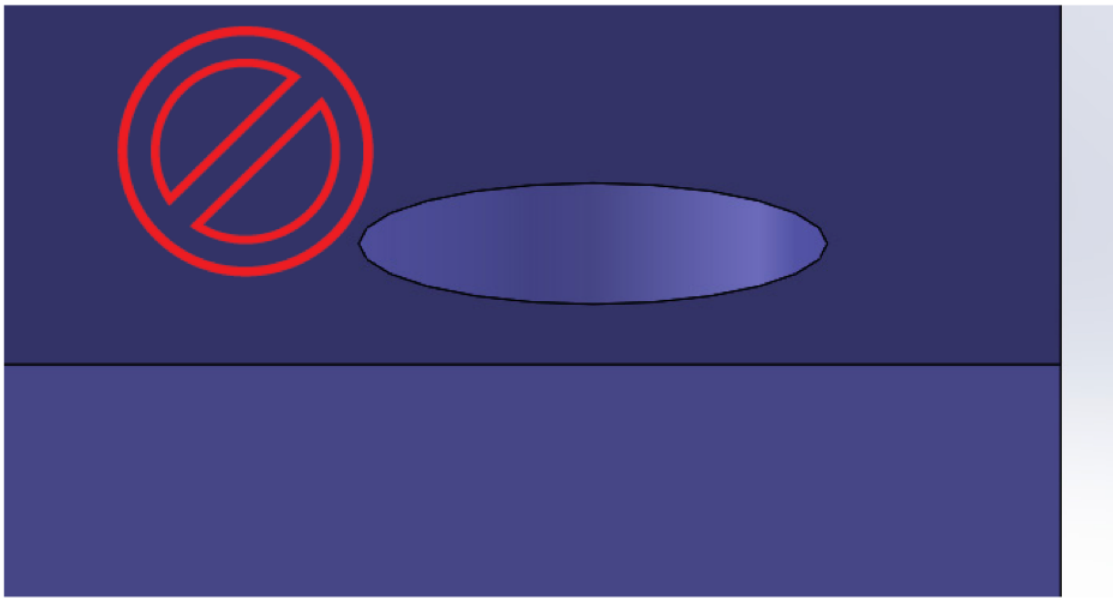
- Threaded holes deeper than 4x the diameter will increase machining time and therefore increase cost. Threaded holes should only be as deep as necessary for your application. Some alternatives include:
 - 1) creating a larger hole
 - 2) creating a through hole and only tapping a short depth from either side
 - 3) tapping a short depth on either side



HOLE DEPTH

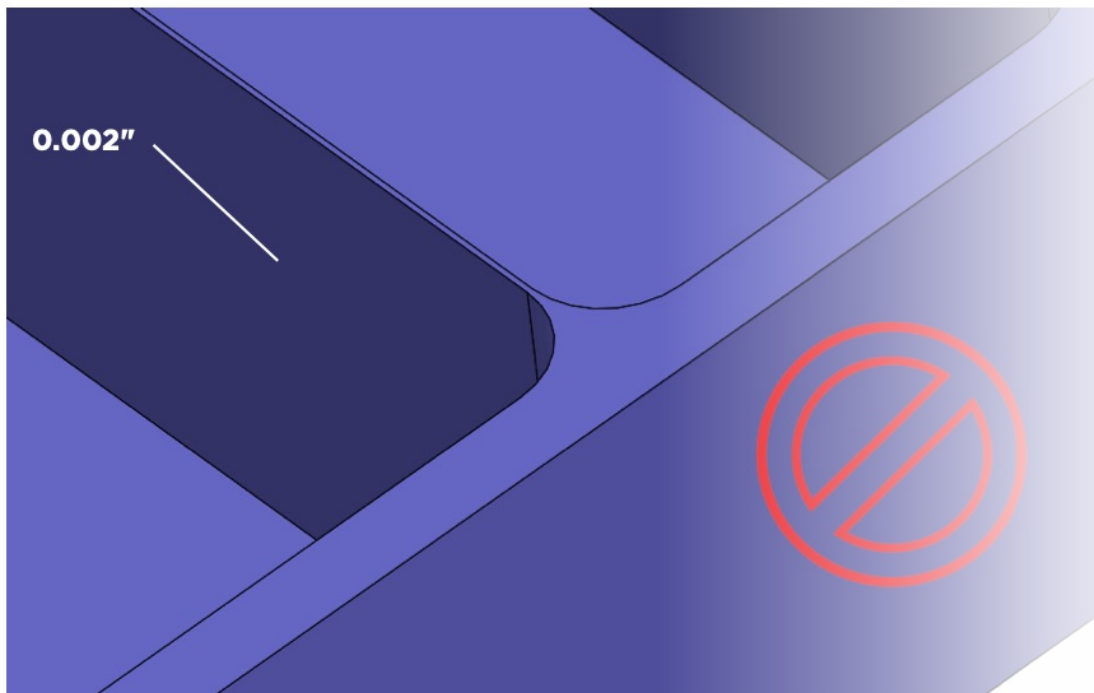
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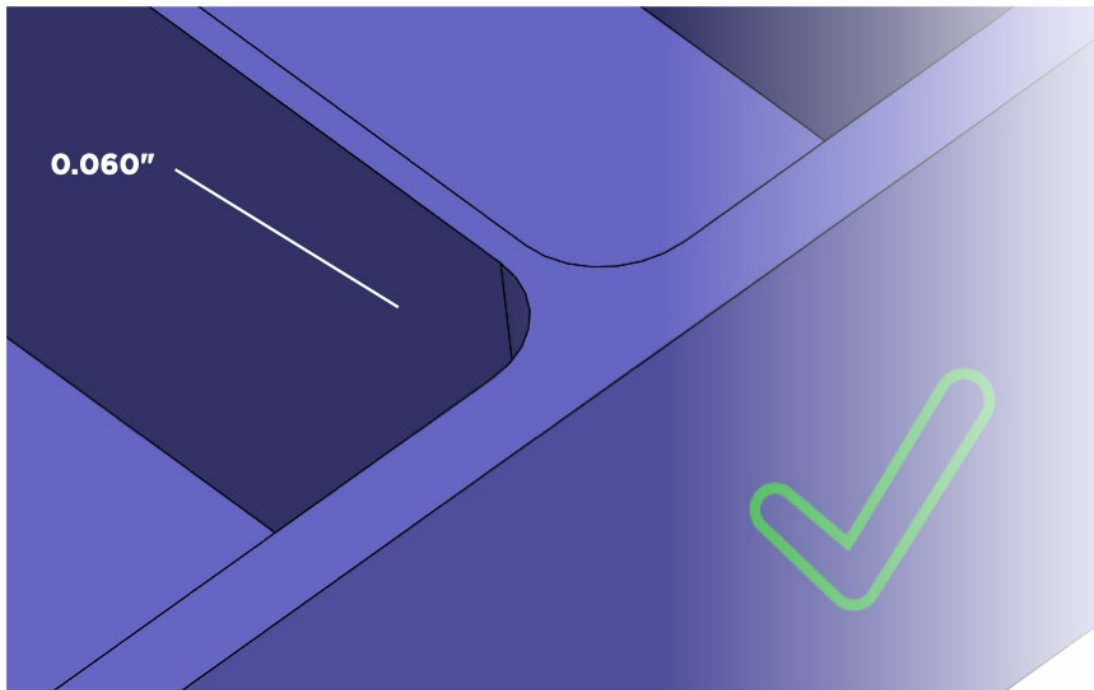
SURFACE FLATNESS

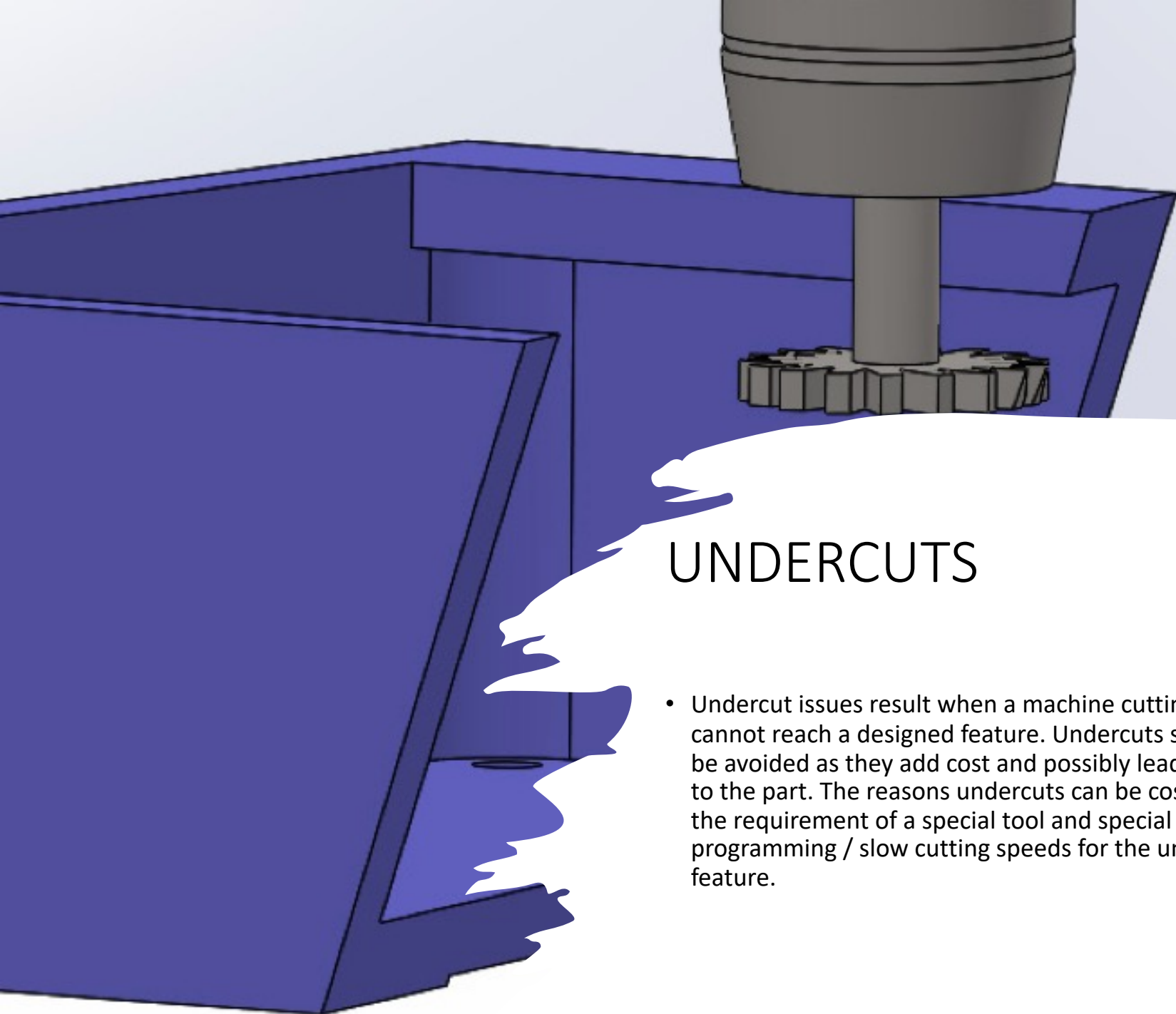
- For machined surfaces with a high degree of flatness, bosses should be used. This clearly defines what areas need to be flatness controlled and simplifies painting and other finishing operations.



WALL THICKNESS

- Machined part wall thickness should be greater than .060". If walls are too thin the part may distort and warp, especially if using a plastic material.

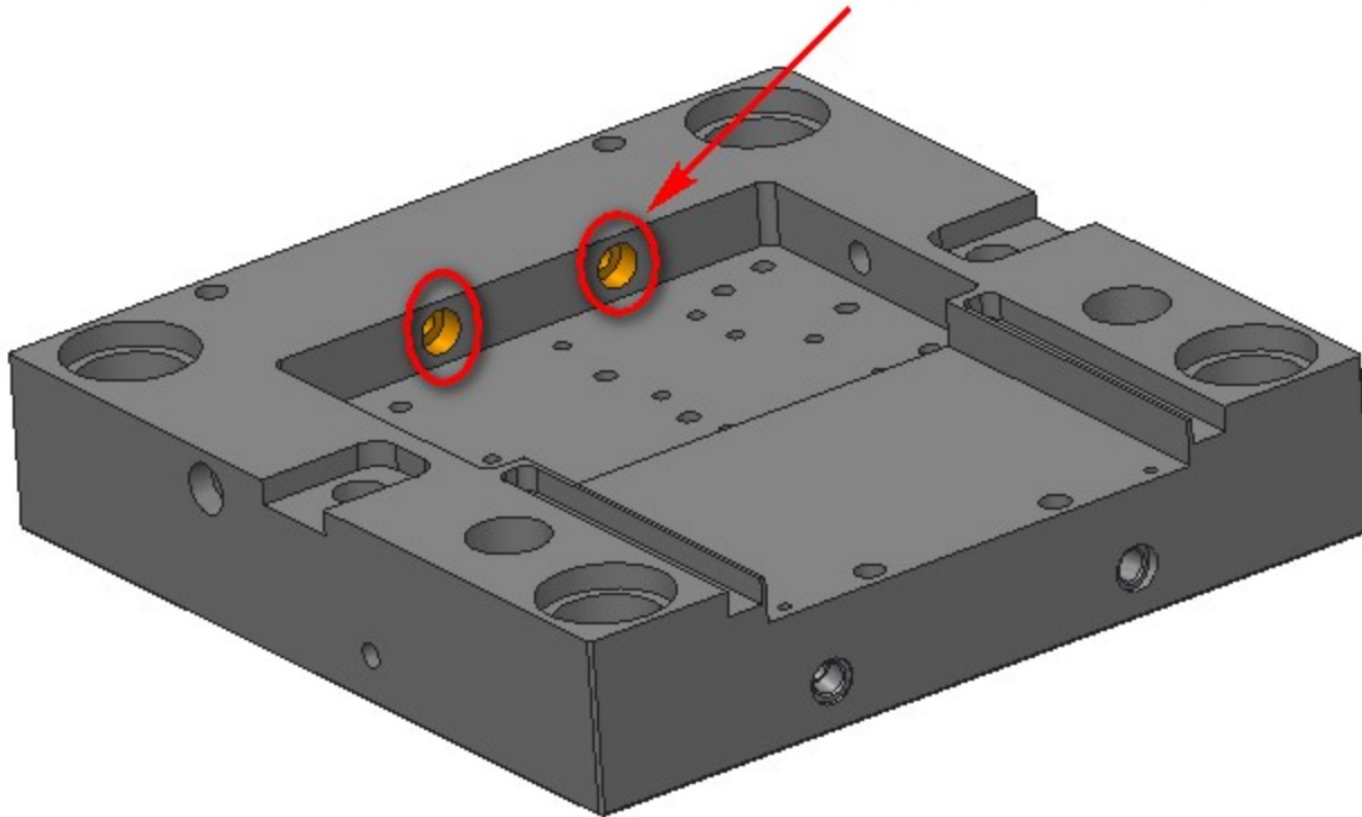




UNDERCUTS

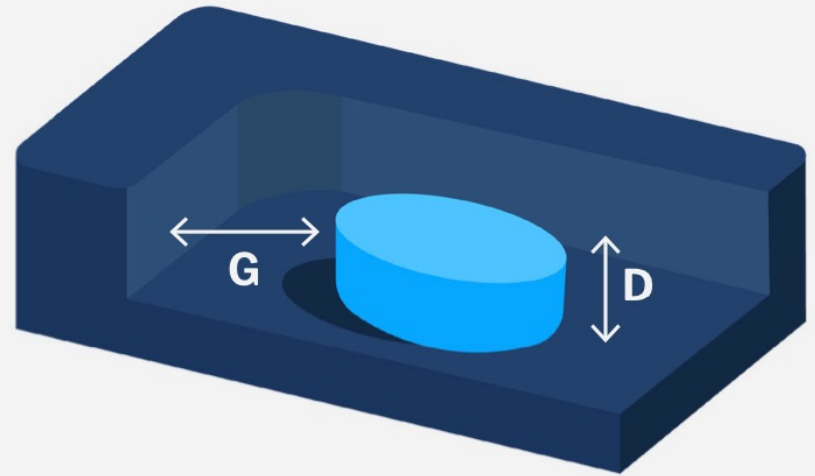
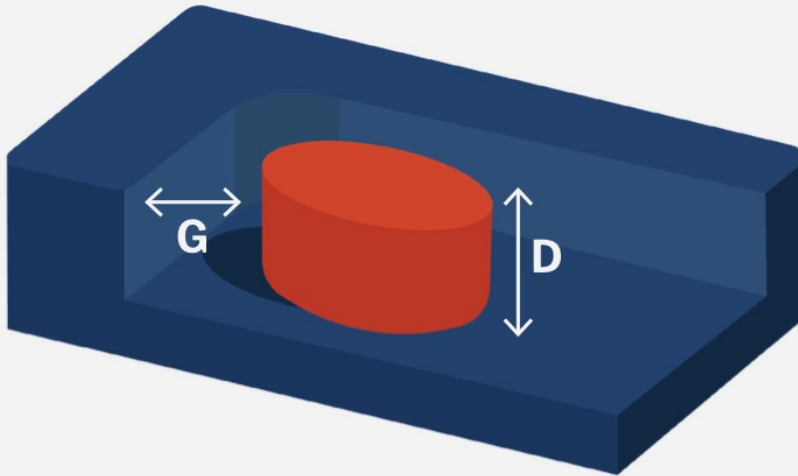
- Undercut issues result when a machine cutting tool cannot reach a designed feature. Undercuts should be avoided as they add cost and possibly lead-time to the part. The reasons undercuts can be costly is the requirement of a special tool and special programming / slow cutting speeds for the undercut feature.

Counterbore hole is opening inside another pocket



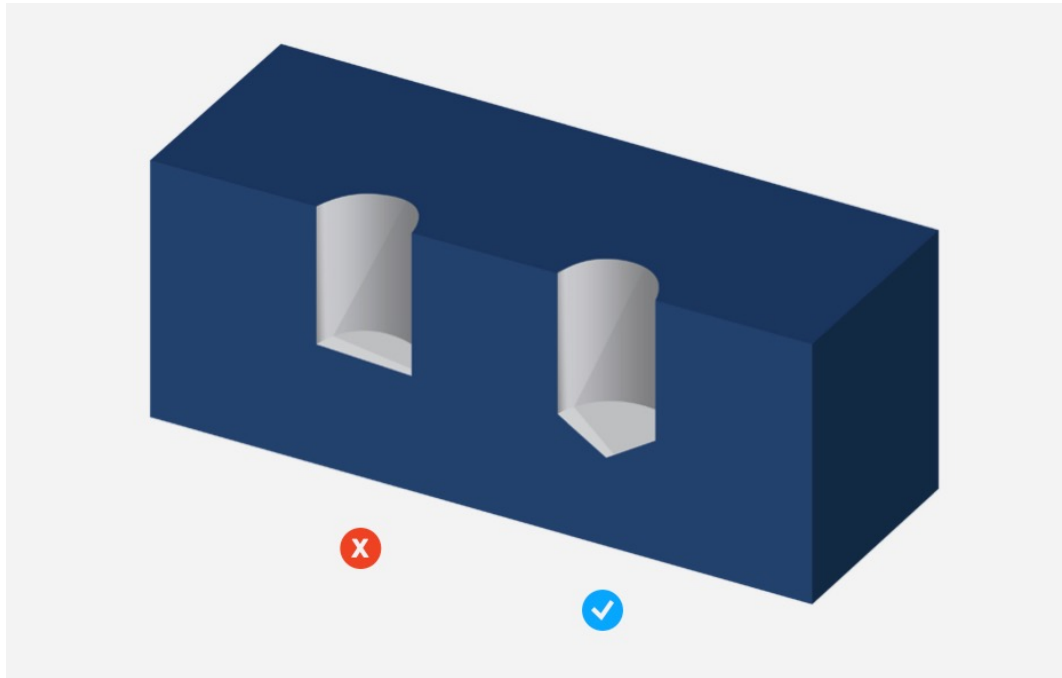
Tool Accessibility

Features should be accessible to the cutting tool in the preferred machining orientation.



Narrow Regions In Pockets

- It is recommended to avoid features that are too close to each other such that the gap between them is too narrow to allow milling cutter to pass through them. If narrow regions are unavoidable, then they should not be very deep. The size of the milling cutter is constrained by the smallest distance between the faces of the feature. Small diameter cutters are prone to breakage and chatter. Hence larger diameter, shorter cutters are generally preferred.



- Flat-bottomed holes require advanced machining operations and often cause problems down the line for subsequent operations like reaming. Avoid creating blind holes with a flat bottom — especially small holes — and instead use a standard twist drill to create holes with cone-shaped bottoms. Cone angles are commonly 118° or 135° .

Drilling

Thank you

